

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology	
Subject: DVD	Aim:	
Experiment No: 9	Date:	Enrollment No: 92400133158

EXPERIMENT 9

AIM: Interactive Dataset Analysis and Dashboard Visualization using PLOTLY

ABOUT DATASET:

This dataset represents students using an **online learning platform similar to Coursera or Udemy**. It analyzes learner demographics, course engagement, study patterns, and learning outcomes. The dataset helps understand **how students interact with online courses, how much time they spend learning, and how engagement affects completion and performance**.

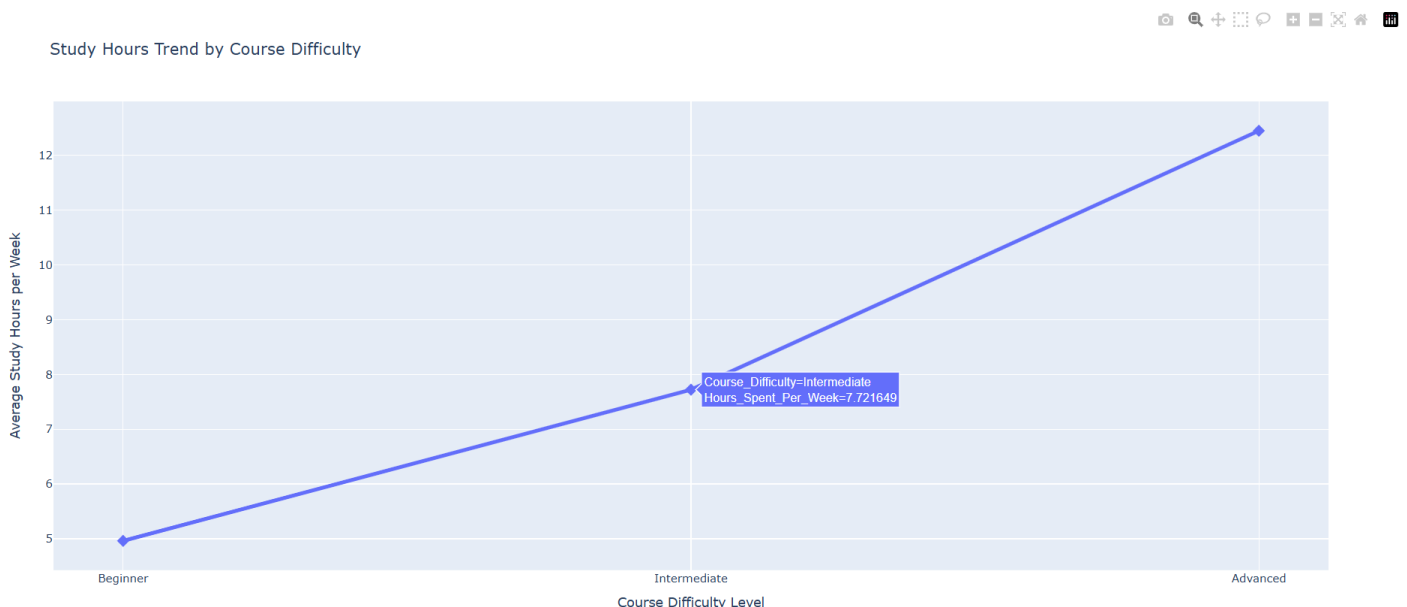
Objectives:

Question 1 – Study Hours Trend by Course Difficulty (Line Chart)

Using Plotly, create a line chart showing the average study hours per week for each course difficulty level (Beginner, Intermediate, Advanced).

Customize the visualization using parameters such as line color, marker style, and line width. Include a title, axis labels, and interactive hover information.

After generating the chart, briefly explain which difficulty level requires the most weekly study time.



As evident from the plot above, “Advanced” as a difficulty level, requires maximum study hours.

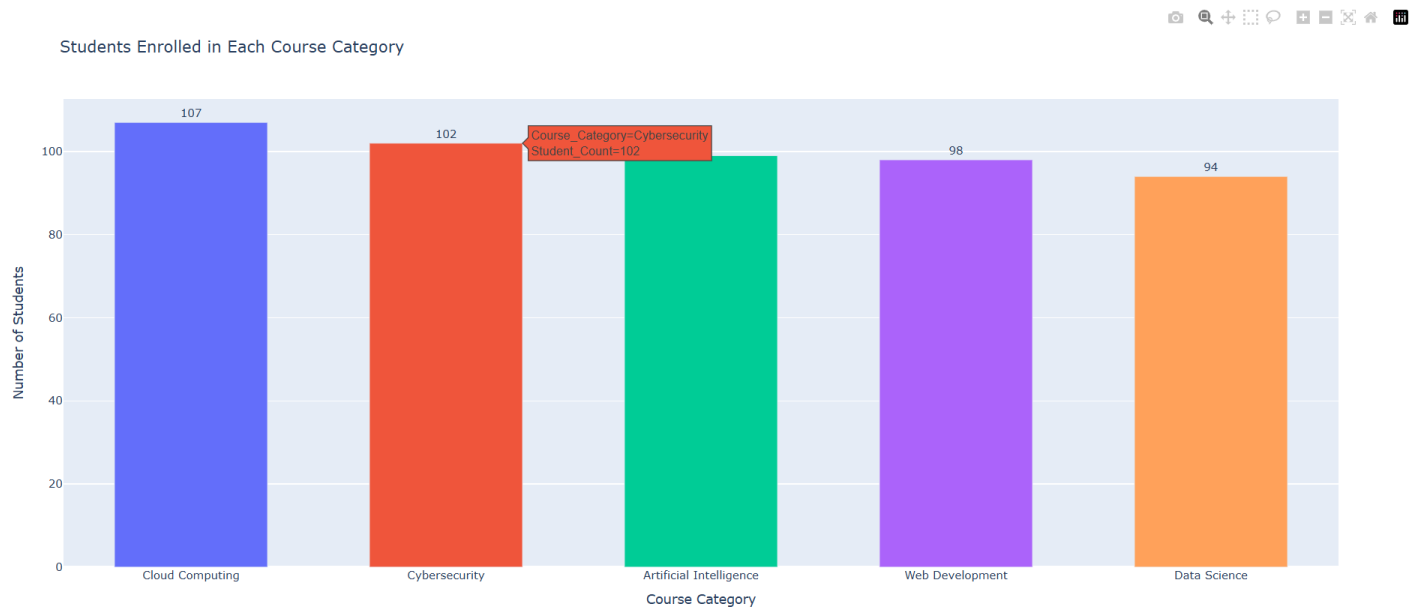
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Question 2 – Students Enrolled in Each Course Category (Bar Chart)

Create a Plotly bar chart showing the number of students enrolled in each course category such as:

- Data Science
- Web Development
- Artificial Intelligence
- Cybersecurity
- Cloud Computing

Customize the chart using different colors, bar width, and text labels showing the number of students. Interpret the visualization and explain which course category is most popular among learners.



Cloud Computing as a Course is very popular among students.

Question 3 – Course Completion Distribution (Pie Chart)

Generate a Plotly pie chart representing the distribution of students based on course completion percentage groups, such as:

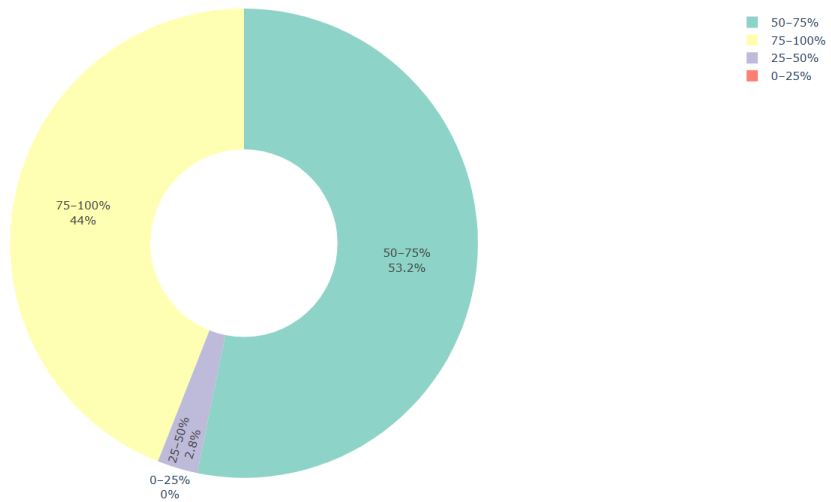
- 0–25%
- 25–50%
- 50–75%
- 75–100%

Customize the visualization using parameters such as color sequence, percentage labels, hole size (donut chart), and hover information.

Explain what the chart reveals about student completion behavior on the platform.

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Student Distribution by Course Completion Percentage



Only 44% students complete 75-100% of the courses they choose.

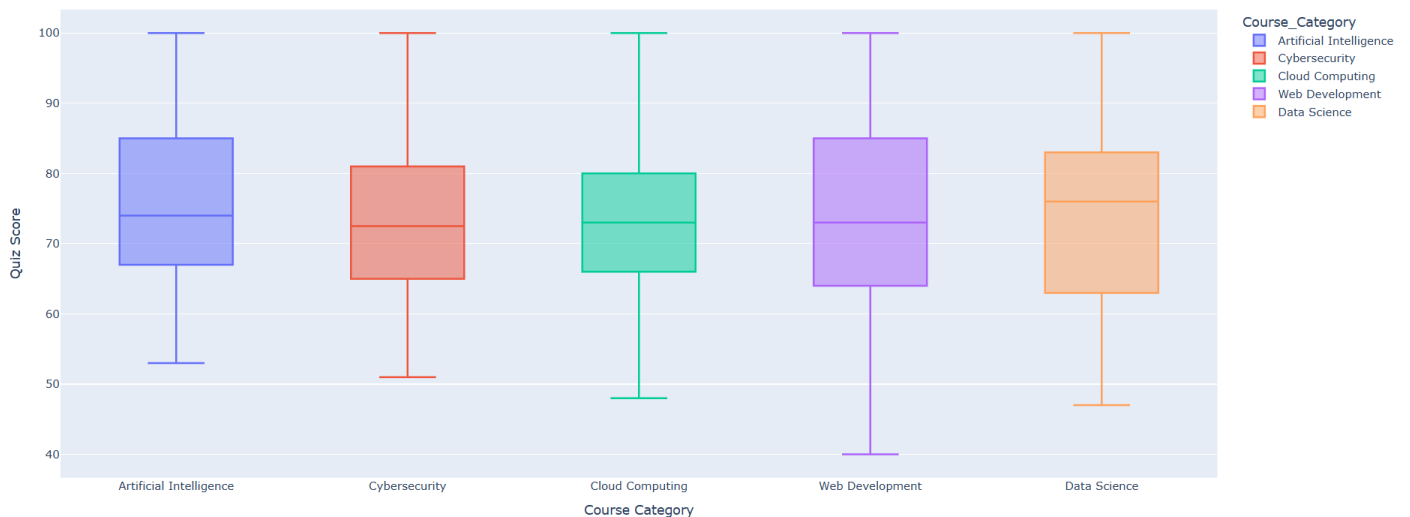
Question 4 – Quiz Performance by Course Category (Box Plot)

Create a Plotly box plot showing the distribution of quiz scores across different course categories.

Modify parameters such as box color, marker color, and outlier display.

Interpret the box plot and explain which course category shows the most consistent quiz performance among students.

Quiz Performance by Course Category



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Cloud Computing shows the most consistent quiz performance among other courses.

Question 5 – Study Hours Comparison by Course Difficulty (Violin Plot)

Construct a Plotly violin plot to visualize the distribution of study hours per week across different course difficulty levels.

Customize the visualization using colors, box visibility, and mean line options.

Analyze the chart and explain whether advanced courses require significantly more study time compared to beginner courses.



Advanced courses require more study time compared to beginner courses.

Question 6 – Learning Engagement Funnel (Funnel Chart)

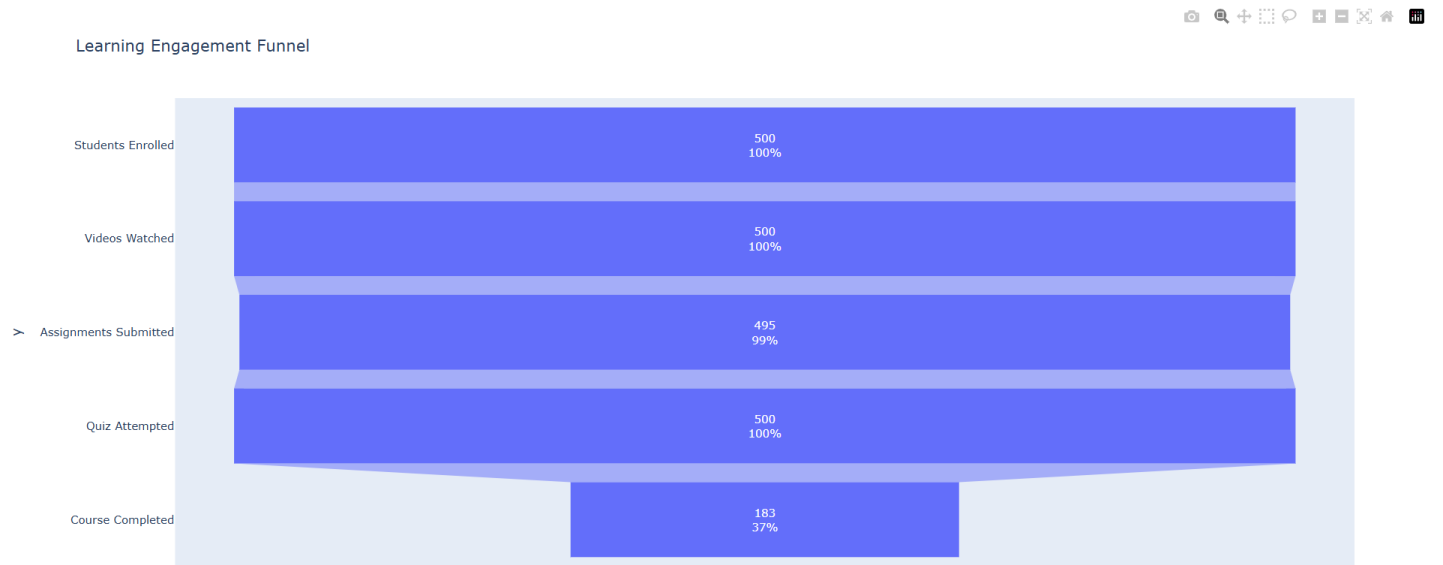
Create a Plotly funnel chart representing the stages of student engagement on the platform:

1. Students Enrolled
2. Videos Watched
3. Assignments Submitted
4. Quiz Attempted
5. Course Completed

Customize the chart using different colors and labels.

Explain how the funnel chart helps identify where student engagement decreases during the learning process.

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Since the Funnel Chart depicts all the steps right from when a student enrolls in the course to the point where they complete it, it's easy to identify the student's engagement through all learning phases.

Question 7 – Coding Practice vs Quiz Score (Scatter Plot)

Create a Plotly scatter plot comparing:

- *Assignments Submitted*
- *Quiz Average Score*

Customize the visualization using parameters such as:

- *Marker size*
- *Marker color*
- *Opacity*
- *Hover labels*

Interpret the plot and explain whether students who submit more assignments tend to achieve higher quiz scores.

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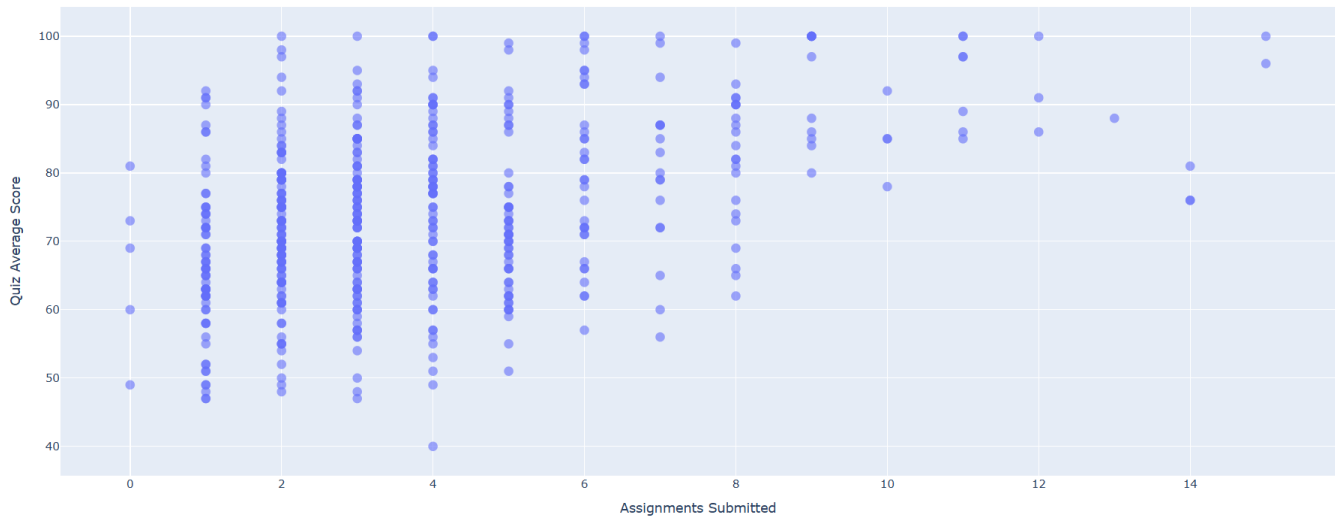









Assignments Submitted vs Quiz Average Score



No as evident from the graph, there is no necessary correlation between number of assignments submitted and scores. Because there are students who have submitted less than 10 assignments yet received a score of 100 in comparison to students who received the same after submitting 15 assignments.

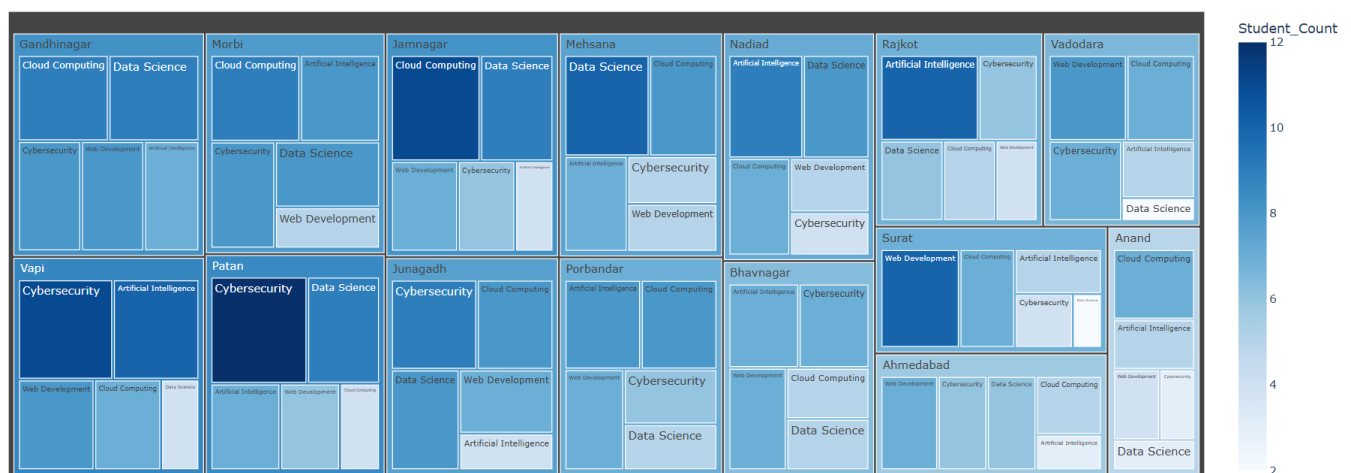
Question 8 – Student Distribution by City and Course Category (Treemap)

Generate a Plotly treemap that visualizes the hierarchical distribution of students by city and course category.

Customize the chart using color scales and labels.

Interpret the visualization and explain which city contributes the highest number of learners in each course category.

Hierarchical Distribution of Students by City and Course Category

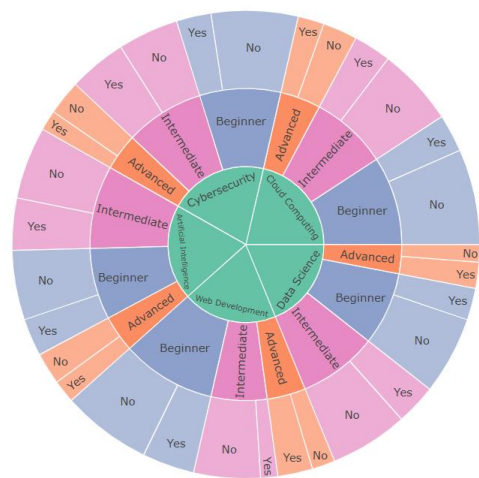


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Patan contributes highest students in Cyber Security, Jamnagar in Cloud Computing, Rajkot in Artificial Intelligence, Surat in Web Development and Mehsana in Data Science.

Question 9 – Learning Activity Hierarchy (Sunburst Chart)
Create a Plotly sunburst chart representing the hierarchy:
Course Category → Course Difficulty → Certificates Earned
Customize the chart using color schemes and labels.
Explain how the sunburst chart helps understand the relationship between course type, difficulty level, and certification outcomes.

Course Category → Difficulty → Certificates Earned



Question 10 – Learning Metrics Correlation (Heatmap)
Generate a Plotly heatmap to visualize correlations between numerical variables such as:

- Hours_Spent_Per_Week
- Videos_Watched
- Assignments_Submitted
- Quiz_Average
- Completion_Percentage
- Learning_Streak_Days

Customize the heatmap using color scales, annotations, and grid lines.
Interpret which variables show the strongest relationships with course completion.

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Correlation Heatmap of Numerical Variables



Videos watched and Hours spent per week come out as the strongest factors for course completion.

Sign:

Date: